About CogSpeed®

CogSpeed is a brief cognitive performance test designed to measure how rapidly a person can identify a simple visual pattern, match it correctly, and continue performing as time pressure increases. The task is built to emphasize processing speed and sustained performance rather than reading, language skill, or memorization.

Core task	**Primary aim**	**Best use**
The center probe shows dots or lines. The subject finds the matching pattern above and taps the response gear in the same position below.	Estimate how well the subject can keep up accurately as presentation pace tightens or stabilizes at a challenging rate.	Repeated self-monitoring, longitudinal tracking, research, field use, and fit-for-duty or readiness screening with appropriate oversight.

Origins and design logic

CogSpeed grew out of earlier adaptive information-processing work developed to measure performance decline under fatigue and demanding operational conditions. It keeps that same core idea in a smartphone-ready form: use a brief, repeatable visual task and adaptive machine pacing to estimate a person's fastest reliable cognitive processing performance.

Rather than trying to measure knowledge, vocabulary, or memory, CogSpeed focuses on how quickly and accurately the subject can perceive, decide, and respond under sustained demand. The design is intended to be portable, repeatable, practical in the lab or field, and broadly usable across cultures because the task relies on simple visual matching rather than language-heavy content.

How CogSpeed adapts

When enough time is available, the matching rule is simple. As the pace becomes more demanding, results increasingly reflect the subject's ability to keep processing correctly and consistently. The adaptive logic is designed to hunt for the fastest presentation rate the subject can reliably manage, while also checking for misses, instability, and other signs that the apparent speed may not represent true sustained performance.

How the app works

The app begins with guided entry steps that can include a saved profile, optional sleep logging, a Samn–Perelli Fatigue Scale (SP-FS) rating, and a seven-page tutorial with direct "Skip to Test" access from every page. The app then runs the selected test mode and saves a result record that can be reviewed immediately in the Speedometer, Results Summary, graphs, and export tools.

Four test modes

- **Mode 1 — CogSpeed Adapted:** starts with calibration and then uses adaptive machine pacing to estimate the subject's blocking level under increasing challenge.
- **Mode 2 — CogSpeed Sustained:** reaches an adaptive level, holds a sustained fixed-rate segment there, and finishes with true self-paced final trials.
- **Mode 3 — Self-paced:** allows rapid repeated responding without machine pacing, useful for practice or alternate tracking needs.

- **Mode 4 — Machine-paced:** uses a calibration-derived fixed pace for a sustained presentation block without adaptive pacing changes.

Workflow and reporting improvements

- **Personal Baseline:** the Speedometer includes an in-app Personal Baseline page that summarizes recent qualifying sessions and helps place a current result in context.
- **Mode 2 CPI/CPA views:** Mode 2 results can be viewed as CPI/MBS or as CPA/Disposition, making it easier to compare speed, stability, and sustained performance quality.
- **Profile Scheduler:** Registered users can select a local on-device scheduler with Anytime, Personal, and Fit-for-Duty reminder options, including sound and voice alerts.
- **Sleep and fatigue context:** optional sleep logging and the Samn–Perelli Fatigue Scale can be saved with each session to improve interpretation of trends over date and time.
- **In-app results review:** the app includes Speedometer, Results Summary, Trial Detail, response graphs, and performance-over-time graphing without requiring an external dashboard.
- **Offline-oriented packaging:** the app is designed as a progressive web app with local storage and service-worker support so it can remain practical in field or low-connectivity use.

How to read the main results

Cognitive Processing Index (CPI) and Maximum Blocking Speed (MBS)

CPI is the main normalized speed summary. MBS is the Maximum Blocking Speed, a pacing-based estimate of the presentation rate at which the subject begins to block. Higher CPI and faster sustainable MBS generally indicate better performance.

Cognitive Performance Ability (CPA) and Disposition

For Mode 2 CogSpeed Sustained, the app also calculates CPA, a broader sustained-performance estimate derived from CPI plus factors such as sustained correct responses, wrongs, misses, variability, drift, lapse rate, recovery ratio, and block efficiency. Zero wrongs can help, zero or very low misses can help, moderate levels can be neutral, and heavier error or instability patterns can lower the result.

> CogSpeed is usually most informative when current results are compared with the subject’s own earlier sessions rather than interpreted as a single isolated number.

Subjective and objective context

CogSpeed pairs objective task performance with the Samn–Perelli Fatigue Scale because self-reported state can add useful context. Subjective decline can appear before larger objective changes are obvious, and repeated use of both measures can help users, researchers, or programs follow trends over time more clearly.

What CogSpeed may be sensitive to

CogSpeed depends on rapid, sustained cognitive throughput. Therefore, it may be sensitive to fatigue, sleep loss, alcohol, some drugs, reduced alertness, normal aging effects, dementia, concussion, other brain injuries, and other conditions that slow or destabilize mental performance. However, CogSpeed does not by itself identify the cause of reduced performance.

Objectivity and anomaly detection

CogSpeed is intended as an objective behavioral screen, but no behavioral test is completely immune to poor effort or unusual response patterns. The app therefore uses timing rules, error limits, repeatability checks, and anomaly-handling logic to reduce artifacts from random tapping, inattention, or other irregular behavior. It is best understood as a rapid, structured estimate of current performance rather than as an all-knowing diagnostic instrument.

Personal, research, and fit-for-duty uses

Personal use	Research use	Fit-for-duty use
Self-monitoring over time, checking whether performance is close to a personal baseline, tracking patterns by sleep, fatigue, time of day, travel, recovery, or aging, and supporting more informed personal decisions about readiness.	Fatigue studies, sleep deprivation studies, circadian rhythm and work-rest research, alcohol and drug studies, concussion and traumatic brain injury studies, gerontology and dementia-related work, and synchronized analysis with electroencephalography or brain imaging.	Operational screening and readiness monitoring for settings where mental alertness is critical, such as transportation, industrial operations, public safety, emergency response, energy, security, and other hazardous work.

CogSpeed can support repeated self-monitoring, research protocols, operational screening, and performance trending when it is used within a defined program. It should not be used as the sole basis for clinical diagnosis, employment action, disciplinary action, or high-stakes operational judgment without surrounding context and appropriate professional review.

Privacy and deployment posture

The app stores its working data locally on the device and is designed to function offline after installation. Profiles, scheduler settings, results history, sleep entries, and review pages are intended to support practical repeated use without requiring constant network access. Deployment-specific policies, backups, exports, and retention should still be governed by the organization or study using the app.

Sponsor and contact

CogSpeed is sponsored by **Gray Matter Metrics, LLC**.

For more information or to provide comments, contact Gray Matter Metrics, LLC: thinkfastgmm@gmail.com

This About page describes the CogSpeed workflow and result structure in plain language. It is not a substitute for study-specific instructions, regulatory materials, or deployment-specific operating procedures.